Project 15: Heart-Disease Risk

Domain-Structured Bayesian Network vs Neural Network

Team T008, Section C | Newton School of Technology, Rishihood University
Nishtha Gupta (2401010303) | Aditya Srivastava (2401010039)
Aanya Mehrotra (2401010006) | Shrijan Sanidhya (2401010449)
Abhay Mallik (2401020002)

Phase 1: four conventional classifiers and a provisional BN
Neural-network implementation belongs to Phase 2

Research question and dataset

How does a provisional BN compare with four ML baselines?

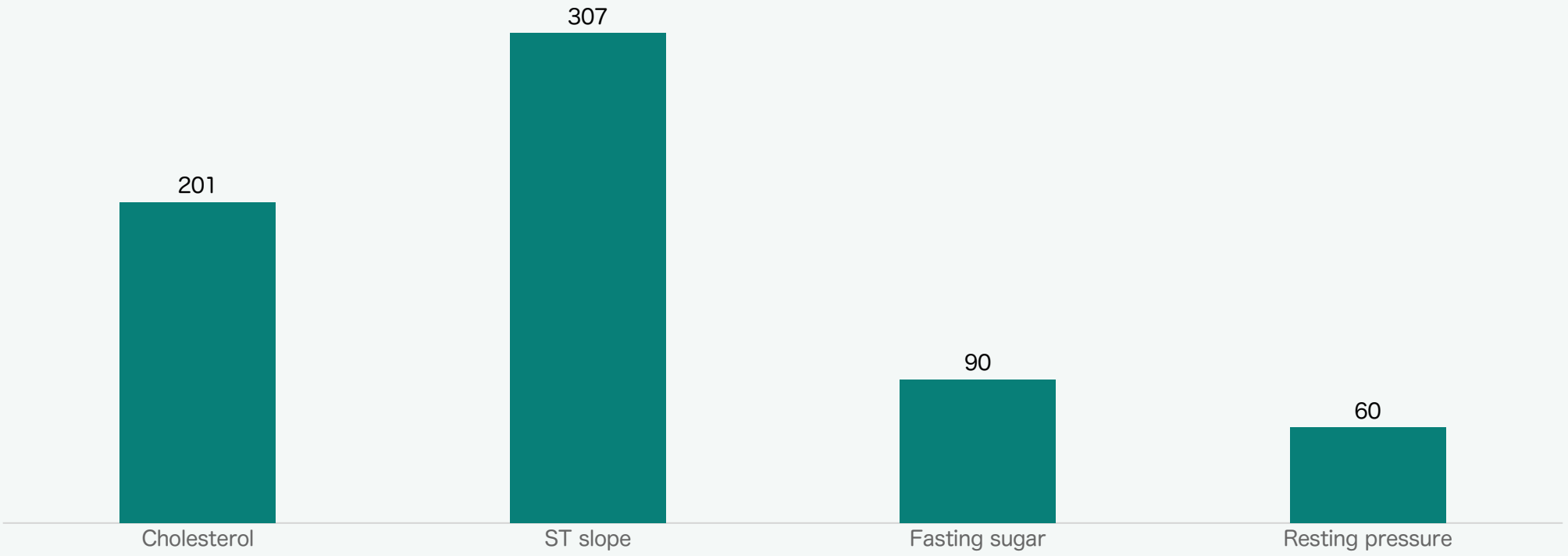
920 rows; 918 after deduplication; 508 positive / 410 negative

ID 304: local Cleveland label differs from its official Hungarian match

Target num > 0; 11 predictors. ID, hospital and target excluded

Source: official UCI processed files checked; exact acquisition/merge history remains incomplete.

Data quality and missingness



Conventional preprocessing fits medians, modes, encoding and scaling inside each training fold.

Counts after zero cleaning, N=918. Full missingness and provenance: report and README.

Selection and retrospective evaluation

Historical split: D=734, T=184. T was already inspected

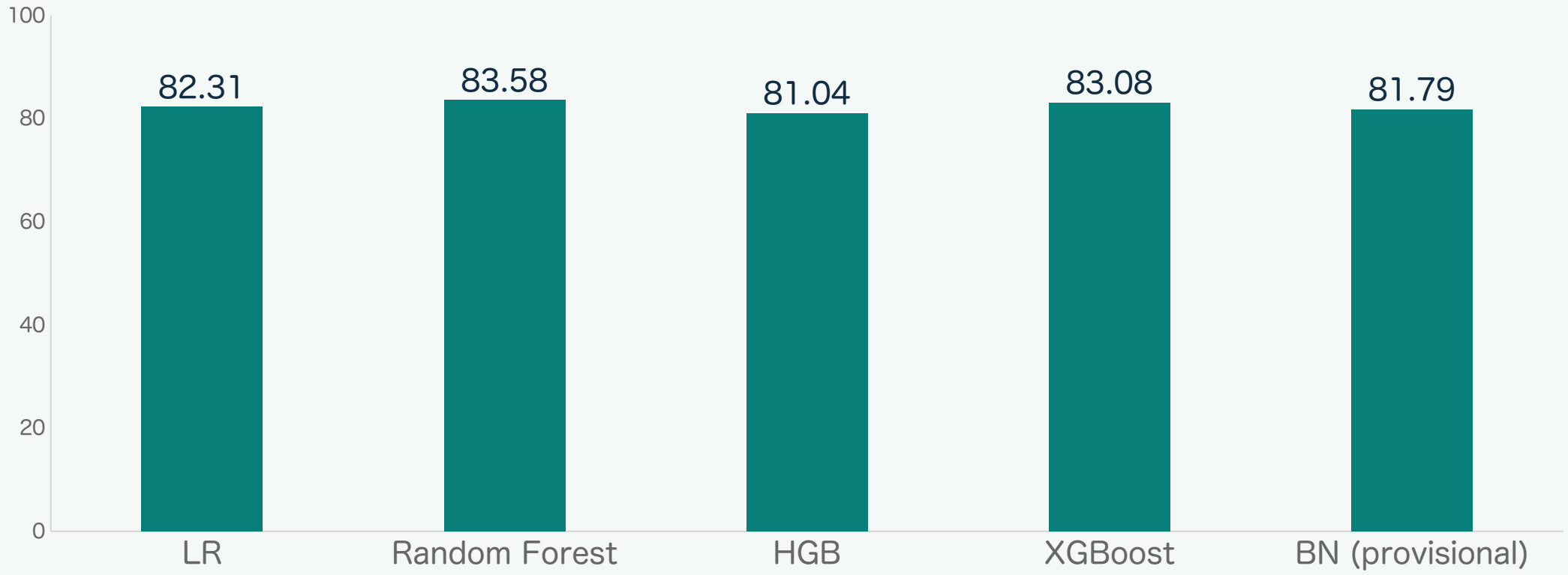
D splits into S=550 selection rows and C=184 cutoff rows

On S: five outer folds, three inner folds. Preprocessing stays local

Frozen S-fitted models predict C; choose cutoffs by F1. No S+C refit

D-only hospital checks repeat family and cutoff selection locally

Nested cross-validation F1 (%)



Random Forest: 0.8358 F1. XGBoost: 0.8308. The predefined rule selects RF.

LR = Logistic Regression. HGB = Histogram Gradient Boosting. Five outer / three inner folds on S.

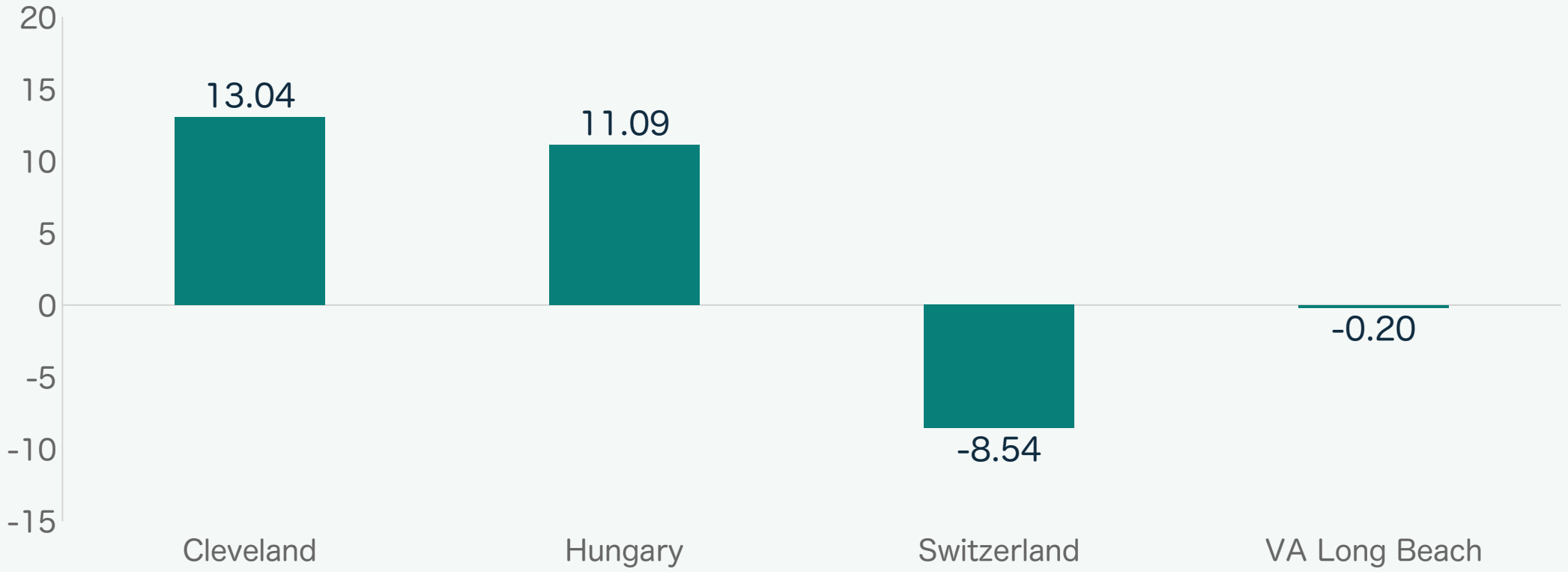
Retrospective historical test evaluation

Model	Cutoff	Accuracy	Recall	F1	ROC-AUC
Logistic regression	0.51	0.8315	0.8824	0.8531	0.9125
Random forest	**0.38**	**0.8043**	**0.9608**	**0.8448**	**0.9029**
HGB	0.37	0.8152	0.9510	0.8509	0.8945
XGBoost	0.34	0.8043	0.9804	0.8475	0.9039
BN (provisional)	0.54	0.7880	0.8137	0.8098	0.8765

RF: 98/102 positives detected; 32/82 negatives falsely flagged (specificity 0.6098)
Cutoff 0.38 is statistical. RF stays selected by nested-CV F1; BN is provisional.

184 previously inspected records. Each model uses its separate C-selected cutoff.

Hospital F1 minus always-positive F1 (pp)



Switzerland: RF F1 0.8712 < reference 0.9565 (88/96 positive)
VA Long Beach: specificity 0.05; only 2/40 negatives correctly rejected

Local family and cutoff selected without the held-out hospital. Zero means equal positive-class F1.

Provisional Bayesian Network

DAG: Y -> all 11 features; oldpeak -> slope (manual provisional graph)

BN theory and UCI ST definitions do not validate these medical edges

CPTs: (count + alpha)/(parent count + K alpha), with alpha = 1

Training-only bins/imputation; exact marginalization of missing evidence

12 edges / 54 omitted pairs reviewed; domain justification remains partial

Reproducibility, limitations and viva

38 tests pass now; all 25 frozen models reproduce saved predictions

Original run: 2940 fits in 340.927 seconds; no new training

App loads saved RF (0.38) and provisional BN comparison (0.54)

Prior test exposure, weak site specificity and ID 304 limit conclusions

Contributions and graph review pending. Neural networks belong to Phase 2